Transcorrelated Jastrow Fitting: Atomization Energies

1 Theory

The procedure described below is motivated by the reference energy correction scheme in TC theory [cite johannes], that states that the basis set error in correlation energy converges quickly with TC, while the errors in the reference energy (i.e., the Hartree-Fock energy) is untouched. To avoid calculations in multiple basis sets and make the workflow more simple and practical, we solve the reference state in a large basis set, and then construct a virtual space with size equal to a lower-level basis. This keeps the reference accurate, while allowing us to perform the correlation calculation in a smaller virtual space that is more similar to the small basis set.

The selected virtual space is constructed by projecting the large-basis virtual MOs onto the small-basis AO space via

$$M = S_{11}^{-1/2} S_{12} C^{\mathrm{virt}} \qquad [n_{\mathrm{small}} \times n_{\mathrm{large,virt}}] \tag{1}$$

where S_{11} and S_{12} are the small-basis and cross-overlap matrices, and C^{virt} collects the large-basis virtual MO coefficients. The elements $M_{\mu a} = \langle \tilde{\varphi}_\mu | \psi_a \rangle$ are overlaps between Löwdin-orthonormalized small AOs and S_{22}-orthonormal large-basis virtuals, so the thin SVD

$$M = U \Sigma V^\top \qquad U\ [n_{\mathrm{small}} \times n_{\mathrm{small}}],\ \Sigma\ [n_{\mathrm{small}} \times n_{\mathrm{small}}],\ V\ [n_{\mathrm{large,virt}} \times n_{\mathrm{small}}] \tag{2}$$

yields singular values $\sigma_k \leq 1$ that are exact cosines of the principal angles between the two subspaces. Dropping the n_{occ} columns of V with near-zero σ_k and projecting onto the AO basis gives the $n_{\mathrm{sel}} = n_{\mathrm{small}} - n_{\mathrm{occ}}$ selected virtuals

$$C^{\mathrm{sel}} = C^{\mathrm{virt}} V_{\mathrm{kept}} \qquad [n_{\mathrm{large,AO}} \times n_{\mathrm{sel}}] \tag{3}$$

which are S_{22}-orthonormal by construction. The SVD rotation mixes canonical virtuals, rendering the projected Fock matrix

$$F^{\mathrm{sel}} = V_{\mathrm{kept}}^\top \,\mathrm{diag}(\varepsilon_a)\, V_{\mathrm{kept}} \qquad [n_{\mathrm{sel}} \times n_{\mathrm{sel}}] \tag{4}$$

non-diagonal. Diagonalizing $F^{\mathrm{sel}} = W \tilde{\Lambda} W^\top$ yields the semicanonical orbitals

$$C^{\mathrm{final}} = C^{\mathrm{virt}}\, V_{\mathrm{kept}}\, W \qquad [n_{\mathrm{large,AO}} \times n_{\mathrm{sel}}]. \tag{5}$$

2 Atomization energies

We compute atomization energies with several xTC method families and benchmark them against SHCI+PBE+CV reference values. All calculations employ effective core potentials (ECPs) for the third-row elements. The SHCI, SHCI+PBE, and conventional CCSD(T) results used here as references are taken from Ref. [2]; the SHCI+PBE energies extrapolated to the complete basis set (CBS) limit are considered the best available benchmark for the G2/55 set. Per-basis, per-molecule method discrepancies and the full basis set convergence of error metrics are hosted as interactive plotly figures [1].

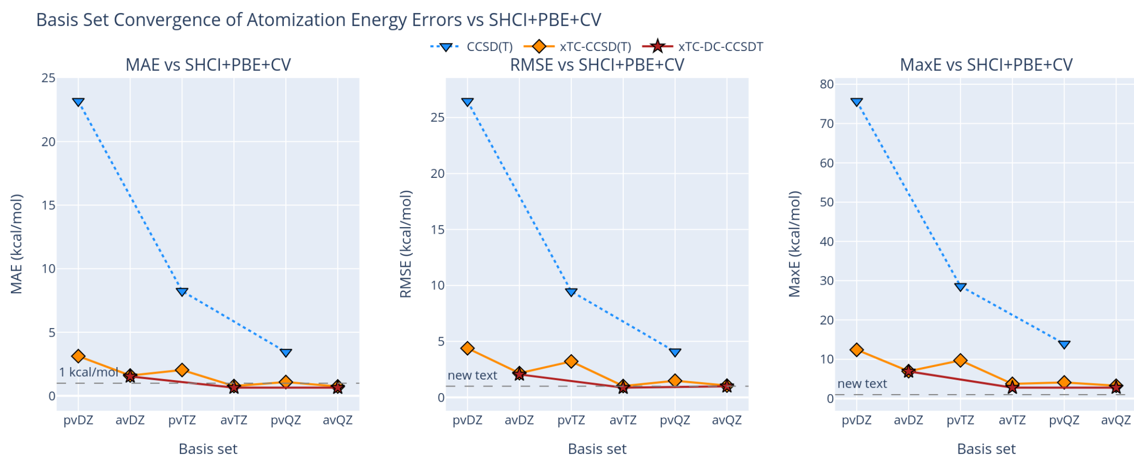

Figure 1: Basis set convergence (MAE, RMSE, MaxE in kcal/mol) of atomization energy errors vs SHCI+PBE+CV for each method family, on a unified x-axis interleaving cc-pVxZ and aug-cc-pVxZ. The interactive version and per-basis per-molecule breakdowns are available online [1].

Table 1: MAE, RMSE, and MaxE of atomization energies vs SHCI+PBE+CV reference (kcal/mol), for each method and basis set.

Method	Basis	MAE	RMSE (kcal/mol)	MaxE
SHCI				
	CBS	0.18	0.26	0.75
xTC-CCSD(T)				
	pvDZ	3.13	4.38	12.40
	avDZ	1.60	2.16	6.95
	pvTZ	2.04	3.21	9.70
	avTZ	0.78	1.01	3.73
	pvQZ	1.09	1.48	4.10
	avQZ	0.74	1.05	3.26
xTC-DC-CCSDT				
	avDZ	1.53	2.05	6.90
	avTZ	0.65	0.87	2.77
	avQZ	0.65	0.99	2.78
CCSD(T)				
	pvDZ	23.18	26.47	75.71
	pvTZ	8.25	9.46	28.60
	pvQZ	3.45	4.08	13.94
	pv5Z	1.67	1.96	5.52

Table 2: Pipeline wall-time breakdown per basis set for the G2/55 set (ECP calculations). RHF and VMC optimisation are performed once at aug-cc-pV5Z and shared across all projection bases.

Basis	Total (nh)	00 RHF (%)	02a+02b VMC opt (%)	00 projection (%)	03 TCHInt (%)	04 CCSD(T) (%)
pvdz	17.2	31.8	33.2	20.6	12.1	2.3
avdz	15.2	36.1	37.6	5.3	18.4	2.6
pvtz	24.4	22.5	23.4	11.4	39.0	3.6
avtz	24.0	22.9	23.9	7.9	41.5	3.9
pvqz	40.4	13.6	14.2	13.9	53.6	4.8
avqz	79.7	6.9	7.2	13.5	66.8	5.7

References

[1] Kristoffer Simula. Interactive plotly figures: xtc atomization energies vs shci+pbe+cv. Online collection, GitHub Pages, 2026. `https://simulak1.github.io/tc-g2-data/`. Pipeline timing waterfall: `https://simulak1.github.io/tc-g2-data/figures/pipeline_waterfall.html`.

[2] Yuan Yao, Emmanuel Giner, Junhao Li, Julien Toulouse, and C. J. Umrigar. Almost exact energies for the Gaussian-2 set with the semistochastic heat-bath configuration interaction method. *J. Chem. Phys.*, 153(12):124117, 2020. doi:10.1063/5.0018577.